Alert-Budgeted Clinical Early Warning: Patient-Blocked Risk Calibration for Stateful Deterioration Alarm Policies

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Abstract

Clinical early warning must balance detection against a manageable alert load, yet threshold-independent discrimination metrics do not specify how often a deployed rule will alarm. We formulate early warning as a stateful alarm policy with a threshold, refractory window and escalation tier, and calibrate it against a budget on the patient-average number of alarms per day. The procedure aggregates within-patient alarm sequences into patient-level losses and applies a finite-grid upper-confidence bound under independent patient sampling, with scorer, policy family and support bound fixed before calibration. A no-alarm fallback is returned when no threshold is certified. We evaluate the procedure on the 40,336 public PhysioNet 2019 records from two hospital systems, using a causal prefix logistic model and a masked GRU over three record-disjoint seeds. On the development site, mean sensitivity rises from 0.185 at a budget of 0.25 alarms per patient-day to 0.774 at 2 per day. Applying the source threshold to the second hospital without target calibration lowers sensitivity at 1 per day from 0.585 to 0.315, even though the realised rate there is lower. An oracle target-site recalibration using target labels raises sensitivity from 0.294 to 0.497 at 1 per day but also raises the realised rate from 0.270 to 0.616; adaptation changes the operating point rather than eliminating the shift. A timestep effective-sample-size comparator is less conservative at tight budgets and increases per-patient budget violations. The results support patient-blocked budget calibration as an auditable engineering instrument, but they do not establish clinical effectiveness, alert-fatigue reduction or deployment readiness.

Keywords: Clinical early warning, Alarm budgets, Stateful policies, Risk calibration, Clinical time series

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1. Introduction

Deterioration events in hospitalised patients unfold over hours, and the physiological trajectory that precedes them is recorded long before the event itself is recognised. This temporal structure has motivated a substantial literature on data-driven early warning, ranging from static severity scores to sequential models that forecast deterioration trajectories from electronic health records [1] and compact, efficiency-aware networks that predict sepsis onset from structured time series [2]. Reported discrimination is high and improving, with in-hospital mortality and onset prediction now routinely exceeding the performance of classical scoring instruments [3]. The implicit promise of this line of work is that a sufficiently accurate model, once installed at the bedside, will shorten time to intervention.

Discriminative prediction and operational alerting address different questions. A score-to-alert conversion requires a decision rule, whereas threshold-independent ranking metrics do not specify how frequently that rule emits alarms. Reviews of intensive-care adverse-event prediction discuss incomplete reporting of alert burden, time-to-alert and deployment calibration [4]. Reviews of multivariate clinical time-series learning also motivate evaluation beyond architectural comparisons [5, 6]. Alert-fatigue studies reach the same conclusion from the workflow side: an alert that is statistically accurate but too frequent or too poorly targeted is dismissed, ignored or actively suppressed [7, 8]. These concerns support reporting the alarm policy and its realised load explicitly, rather than treating an area-under-the-curve comparison as a specification of bedside alerting behaviour.

Recent research improves the backbone that produces the risk score. For irregularly sampled and heterogeneously missing observations, approaches include reliability-aware temporal aggregation [9, 10], graph formulations over irregular observations [11], and masked-attention models for deterioration prediction [12]. Hybrid sequence-graph models report strong discrimination on the 2019 cohort [13], structure-preserving models attempt to keep each alert connected to its evidence [14], and operational early-warning indices are being developed outside the public benchmark [15]. Transfer-oriented variants align feature spaces across cohorts [16], while dynamic temporal graphs incorporate clinical calculators into the prediction pathway [17]. These methods motivate prospective backbone comparisons, but their adapters are not included in the current reference implementation. Our implemented contribution instead concerns the causal mapping from fixed scores to stateful

alarms and the risk associated with selecting that mapping.

A separate literature attaches reliability estimates to clinical predictions. Selective prediction and calibration have been analysed for multimodal clinical classification [18], reliability estimation has been extended beyond output-space calibration for time-series classifiers [19], and conformal methods have been used for physiological regression uncertainty [20]. Conformal risk control extends coverage to a user-specified expected loss [21], and recent work relaxes its monotonicity requirements and studies cross-validation variants [22, 23]. Time-series conformal methods also model temporal dependence directly [24, 25]. Our distinction is the unit and estimand of alarm-policy calibration: refractory state makes successive emitted alarms dependent, so those hours cannot simply be counted as independent patient samples. This does not imply that every conformal treatment of time series is invalid or that a stateful policy cannot be calibrated at the trajectory level. The reference construction uses independent patient blocks and fixed finite-grid upper-confidence bounds, a stronger sampling assumption than exchangeability alone.

This paper makes the alarm budget an explicit constraint and calibrates the policy that generates alerts. The implementation separates causal score estimation from full-trajectory policy replay. Its finite-sample bound concerns the population mean of patient-level rates, not a hard maximum for each patient or ward. We evaluate the procedure on the public 2019 A/B cohort with three record-disjoint seeds, including a same-threshold external audit and a deliberately invalid timestep comparator. The contributions are as follows.

- We formulate early warning as a stateful alarm policy in which the refractory window and escalation tier are control variables, so that alarm volume becomes a design parameter of the monitoring system rather than an emergent property of a threshold.
- We derive a patient-blocked finite-grid upper-confidence bound on the population alarm rate under independent sampling, and distinguish it from a deliberately invalid timestep effective-sample-size shortcut.
- We provide causal reference learners, official-format data readers and cached-score experiment drivers, and report a real-data evaluation under shared requested budgets, including a cross-hospital transport audit and an explicit negative result for architecture-only claims.

The remainder of the paper reviews related work (Section 2), develops the policy formulation and calibration procedure (Section 3), specifies the datasets and experimental protocol (Section 4), reports the results (Section 5), and discusses their scope and limitations (Section 6).

2. Related Work

2.1. Backbones for Irregularly Sampled Clinical Time Series

Recent work models irregular sampling explicitly, using continuous-time formulations [10], reliability-aware temporal aggregation [9], and graphs over observations and variables [11]. Hybrid sequence-graph designs report strong discrimination on the 2019 cohort [13], while structure-preserving sequence models attempt to keep each alert connected to its supporting evidence [14]. Complementary approaches model the observation process to synthesise irregular trajectories [26] or exploit informative missingness for ward deterioration prediction [12]. Domain-specific applications include venous thromboembolism detection after cancer treatment [27], while surveys synthesise the wider modelling landscape [28]. These methods are potential future score backbones, not models already integrated or reproduced in the current package. Any integration must retain prefix-only information and cannot reuse a full-trajectory imputation procedure without a separate causality audit.

2.2. Early Warning and Deterioration Prediction

Clinical early warning has moved from static severity scores toward dynamic, trajectory-aware prediction. Classical early-warning scores remain important baselines and have been externally validated in emergency and prehospital settings [29, 30], but they are typically fixed-weight instruments rather than calibrated policies. Recent multicentre work predicts the shape of sepsis deterioration rather than a single risk estimate, arguing that heterogeneous trajectories require patient-specific forecasting [1], while efficiency-oriented designs pursue compact models that remain explainable under the compute constraints of a ward workstation [2]. Others inject clinical reasoning into the prediction pathway: dynamic temporal graphs embed calculator-derived features so that the model inherits the semantics of established instruments [17], and language-model-guided simulation is used to generate interpretable physiological dynamics for early warning [31]. Transfer-oriented formulations address the practical reality that a model trained in one institution must be reused in another with different coding practice and case mix

[16], and earlier work established that deep models over longitudinal inpatient records can outperform tabular baselines for outcome prediction [3, 32, 33]. Prospective implementation studies show that deployment changes the evidentiary question: a model that performs well offline may produce modest clinical effects and a substantial alert burden in practice [34]. Multisite and external validations of sepsis models likewise report performance degradation and calibration drift across hospitals [35, 36]. Across this literature the model is usually the unit of contribution; the alerting rule that converts its output into an operational signal is treated as an afterthought, and comparison between methods is almost always conducted at a fixed threshold rather than at a fixed alarm volume.

2.3. Calibration, Selective Prediction, and Conformal Guarantees

Calibration and selective prediction have been studied in multimodal clinical classification [18]. Reliability estimation for time-series classifiers also examines evidence beyond output remapping [19], while adaptation audits consider realistic label delays [37]. Conformal risk control provides distribution-free control of an expected loss and has been extended to non-monotone losses and cross-validation variants [21–23]. Time-series conformal methods handle temporal dependence explicitly [24, 25, 38], and prediction-set design has been studied for efficiency and human decision quality rather than coverage alone [39, 40]. Conformal methods have also been applied to physiological regression uncertainty [20], to performance estimation under distribution shift [41], and to patient-specific haemodynamic adaptation [42]. These contributions concern different units, targets and sampling assumptions; their guarantees cannot be declared invalid merely because an unrelated downstream alarm rule has memory. The present reference procedure instead specifies a patient-average alarm-rate estimand and uses independent-block concentration, without claiming the weaker assumptions or different risk statement of standard conformal constructions.

2.4. Evaluation Practice

Risk-of-bias instruments for prediction models address artificial-intelligence-specific issues in calibration, heterogeneity and applicability [43]. Reviews of intensive-care adverse-event and time-series prediction also discuss reporting and evaluation limitations [4–6]. The alert-fatigue literature adds a complementary measurement problem: dismissal, override and workflow burden are not captured by discrimination or calibration

[7, 8, 44]. Data-sharing histories and foundation-model agendas also shape which cohorts are available for replication [45], and task heterogeneity across diagnoses complicates direct transfer of mortality models [46, 47]. These concerns motivate a protocol that reports requested budgets, realised alarm rates, sensitivity and lead time alongside discrimination. The present evaluation follows that protocol on the public 2019 cohort; shared requested budgets make the operational constraint explicit but do not imply identical realised alert volume across models.

3. Method

3.1. Problem Setting and Notation

Let patient i contribute observations $\mathcal{O}_i = \{(t_{ij}, m_{ij}, v_{ij})\}_{j=1}^{n_i}$, with observation times t_{ij}, variable indices m_{ij}, and values v_{ij}. The number and spacing of observations may differ across patients and variables. Decisions are made on a prespecified regular grid with sampling interval Δ hours, using only observations available at each decision time. A causal score function maps this history to $s_{it} = f_\theta(\mathcal{O}_{i,\le t}) \in [0, 1]$; any preprocessing, including imputation, must satisfy the same restriction.

An observed event has time $t_i^\star$, and a useful early alarm falls in $[t_i^\star - W, t_i^\star)$, where $W = 12$ hours is a design choice for the proposed evaluation. Records without an observed event are represented separately from records whose event time is outside observed follow-up. Clinical event time and the boundary of a shifted prediction label are not interchangeable. The data protocol in Section 4 therefore requires an explicit onset convention and does not evaluate lead time against an unobserved or left-censored clinical onset.

3.2. Stateful Alarm Policies

The policy $\pi_\lambda = (\lambda, \tau, \kappa)$ comprises a threshold, an integer refractory duration, and an integer escalation duration. Here $\tau \ge 0$ is the number of subsequent decision steps suppressed after an alarm; $\kappa = 0$ disables escalation, while $\kappa > 0$ permits an override after κ consecutive above-threshold steps following the last alarm. Let z_{it} be the number of refractory steps remaining at the start of step t, and let c_{it} include the current exceedance and the consecutive suppressed exceedances immediately preceding it. With

initial state zero, the emission and update are

$$a_{it} = \mathbf{1}[s_{it} \text{ finite},\ s_{it} \geq \lambda,\ z_{it} = 0\ \vee\ (\kappa > 0\ \wedge\ c_{it} \geq \kappa)]\,,$$
$$z_{i,t+1} = \begin{cases} \tau, & a_{it} = 1, \\ \max(0, z_{it} - 1), & a_{it} = 0. \end{cases} \tag{1}$$

The exceedance counter resets after an alarm or a non-exceedance, including a missing score. The refractory duration counts decision steps, not observations in the irregular raw record. Setting $\tau = 0$ gives the memoryless threshold policy, whereas $\lambda = +\infty$ gives an exact no-alarm policy. The latter is an explicit safe fallback and is distinct from the largest finite grid threshold, which can still emit alarms.

The scorer and structural parameters are selected using training and validation patients and are then fixed before risk calibration. Every candidate threshold is evaluated by replaying the entire causal policy, with state reset at each patient's admission. Within-patient alarm indicators are dependent, so counting them as independent calibration units is not justified. Dependence within a patient does not itself prevent calibration over independently sampled patient trajectories.

3.3. Risk Estimand and the Timestep Shortcut

For n_i decision steps, define exposure $D_i = n_i\Delta/24$ days and alarm count $A_i(\lambda) = \sum_t a_{it}(\lambda)$. The target is the patient-average daily alarm rate

$$R(\lambda) = \mathbb{E}[A_i(\lambda)/D_i]\,. \tag{2}$$

This estimand weights each patient equally and is not the pooled ward rate $\mathbb{E}[A_i]/\mathbb{E}[D_i]$, which weights patients by exposure. Both may be clinically relevant, but their identification requires different loss definitions. The present certificate concerns only Equation (2); it does not guarantee a fixed maximum count for every patient or a fixed realised total for a ward.

The synthetic comparison includes a deliberately invalid effective-sample-size shortcut. It uses the same empirical patient-average losses as the blocked selector, but substitutes the number of calibration timesteps for the number of calibration patients in the confidence radius. This isolates the consequence of inflating the unit of information without changing the point estimator. It is not an implementation of a particular published timestep-conformal algorithm, nor does the comparison establish that every alternative time-series calibration method is invalid.

3.4. Patient-Blocked Uniform Risk Bounds

Define the patient loss as the observed daily alarm rate,

$$L_i(\lambda) = A_i(\lambda)/D_i \in [0, L_{\max}]. \tag{3}$$

The upper bound must hold over the deployment population's support, not merely over the observed calibration sample. If eligible records have at least $H_{\min}$ hours of exposure, let $n_{\min} = \lceil H_{\min}/\Delta \rceil$ and $g = \tau + 1$ without escalation, or $g = \min(\tau + 1, \kappa)$ when escalation is enabled. The minimum firing separation is g, so the exact discrete-support upper bound is

$$L_{\max} = \sup_{n \in \mathbb{N},\ n \geq n_{\min}} \frac{24}{n\Delta} \left(1 + \left\lfloor \frac{n-1}{g} \right\rfloor \right). \tag{4}$$

The rate decreases between firing-count increments, and its peaks at $n = 1 + qg$ decrease as q increases; hence checking $n_{\min}$ and the next such peak is sufficient. For hourly sampling, $\tau = 4$, no escalation and $H_{\min} = 24$, a 26-hour stay yields $L_{\max} = 144/26$, exceeding the five alarms per day possible over exactly 24 hours. Escalation can increase this bound and must not be omitted from its calculation. Losses outside the declared interval are rejected, not clipped, because clipping would certify a different estimand.

Condition on independently obtained training and validation data. The scorer, structural parameters, finite threshold grid Λ, and support bound are fixed under this conditioning. Calibration trajectories are independent and identically distributed draws from the same population as future deployment trajectories, although each trajectory may have arbitrary internal dependence. Exchangeability alone is not sufficient for the independent-sample concentration argument used here.

Proposition 1 (Budget validity)**.** *Let* $\widehat{R}_n(\lambda) = n^{-1} \sum_{i=1}^{n} L_i(\lambda)$ *and* $\alpha \in (0, 1)$*. Under the conditions above, select*

$$\widehat{\lambda} = \min \left\{ \lambda \in \Lambda : \widehat{R}_n(\lambda) + L_{\max} \sqrt{\frac{\log(|\Lambda|/\alpha)}{2n}} \leq B \right\} \tag{5}$$

when the set is nonempty, and select the exact no-alarm policy $\widehat{\lambda} = +\infty$ *otherwise. For a finite nonnegative budget* B*, the selected policy satisfies*

$$\Pr\{R(\widehat{\lambda}) > B\} \leq \alpha. \tag{6}$$

The probability is over calibration samples; R *is a population expectation conditional on the fixed scorer and policy family.*

Proof. For each fixed λ, the relevant one-sided Hoeffding inequality is

$$\Pr\{R(\lambda) - \widehat{R}_n(\lambda) > \epsilon\} \leq \exp(-2n\epsilon^2/L_{\max}^2).$$

Setting $\epsilon = L_{\max}\sqrt{\log(|\Lambda|/\alpha)/(2n)}$ and applying a union bound gives simultaneous upper bounds $R(\lambda) \leq \widehat{R}_n(\lambda) + \epsilon$ for all grid policies with probability at least $1 - \alpha$. On that event, any policy selected from the feasible set satisfies $R(\widehat{\lambda}) \leq B$. If the feasible set is empty, the fallback has identically zero loss and hence satisfies the budget without a concentration argument. Thus a budget violation can occur only on the failure event of the simultaneous upper bounds, which has probability at most α. □

For $n \geq 2$, the implementation also permits the empirical-Bernstein radius[1]

$$\epsilon_{\mathrm{EB}}(\lambda) = \widehat{\sigma}(\lambda)\sqrt{\frac{2\log(2|\Lambda|/\alpha)}{n}} + \frac{7L_{\max}\log(2|\Lambda|/\alpha)}{3(n-1)}, \tag{7}$$

where $\widehat{\sigma}$ is the sample standard deviation with denominator $n - 1$. This bound can improve on Hoeffding when variance is small and the sample is sufficiently large, but it is not uniformly tighter. Both variants are finite-grid upper-confidence risk procedures; patient blocking does not turn either into the standard expected-risk conformal construction. The manuscript's conformal terminology denotes this broader calibration motivation, not a claim that the proof requires only conformal exchangeability.

The certificate always uses the raw empirical loss at the deployed threshold. Stateful alarm counts are not assumed to be monotone in the threshold, so isotonic regression is optional descriptive output and never substitutes for that loss in the upper bound. A finite test-cohort mean can exceed the budget even when its population expectation does not, so its exceedance frequency is an empirical diagnostic rather than the exact probability in Equation (6). Likewise, calibration transferred across distributions is not covered without the same-population assumption.

3.5. Operating Points and Lead-Time Evaluation

The reference selector deploys the smallest finite threshold whose raw risk upper bound meets the requested budget. This is a deterministic operating-

[1]The finite-class empirical-Bernstein bound and its constants are given by Maurer and Pontil, *Empirical Bernstein Bounds and Sample Variance Penalization*, https://arxiv.org/abs/0907.3740.

point convention rather than a guarantee of optimal detection. A separate utility objective could select among the uniformly certified candidates without invalidating the risk bound, but such an optimiser is not implemented here. Neither a lower threshold nor a higher budget is assumed to improve every operational metric of a stateful policy.

For an event observed during follow-up, define the detection window $\mathcal{W}_i = [\max(0, t_i^\star - W), t_i^\star)$. An event is detected if at least one alarm falls in that window, and its lead time is

$$\ell_i(\lambda) = t_i^\star - \min\{t \in \mathcal{W}_i : a_{it}(\lambda) = 1\}. \tag{8}$$

Lead time is undefined for undetected events and is represented as missing rather than zero. The reported median therefore describes detected events only and must always be accompanied by sensitivity and the number of eligible events. Events at admission, outside observed follow-up, or without an identifiable onset cannot supply an observed pre-onset lead time. A retrospective top-k selection over a complete trajectory is noncausal and may serve only as a labelled oracle comparator, not as a deployable early-warning policy.

Budget sweeps compare methods under the same requested upper-risk constraint, not necessarily at identical realised alarm rates. The plotted frontier uses realised alarm rate and conditional median lead time, retaining only nondominated pairs. At equal alarm rates only the largest lead time is retained; at larger rates a point remains only when its lead time strictly improves. Because conditional lead time can improve by missing difficult cases, sensitivity must be reported alongside the frontier before any clinical interpretation.

4. Experimental Setup

4.1. Data Sources and Cohort Audit

Figure 1 summarizes the evaluation design. We use the public PhysioNet/CinC Challenge 2019 cohort [48], which contains 20,336 records from site A and 20,000 records from site B. The two sites are treated as distinct hospitals rather than interchangeable shards. All records were downloaded from the public release and checked against the published object sizes and MD5 entities before parsing. Site A contains 1,790 records with a positive sepsis label and 203 left-censored records whose first observed label is already

Algorithm 1 Patient-blocked upper-confidence calibration of an alarm policy

Require: independent calibration patients; fixed scorer, grid Λ, structural parameters, valid $L_{\max}$; budget B; level α

for $\lambda \in \Lambda$ **do**
    replay the causal policy on each patient's observed follow-up
    compute $L_i(\lambda)$ and reject any loss outside $[0, L_{\max}]$
    compute raw $\widehat{R}_n(\lambda)$ and Hoeffding or empirical-Bernstein radius
end for
$\mathcal{F} \leftarrow \{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon(\lambda) \leq B\}$
$\widehat{\lambda} \leftarrow \min \mathcal{F}$ if nonempty; otherwise $+\infty$
return threshold, raw certificate, and explicit fallback status

positive; site B contains 1,142 positive and 223 left-censored records. After the six-hour label shift described below, 1,587 A records and 919 B records have an identifiable onset inside observed follow-up. Median observed stay is 39 hours at A and 38 hours at B.

The audit also exposes substantial measurement heterogeneity. In site A, $EtCO_2$ is absent in every record, whereas 7.6% of B records contain at least one $EtCO_2$ value; temperature is observed in only 33.8% of A records, and several laboratory variables are observed in fewer than 5%. This pattern motivates retaining a fixed variable set across sites and explicitly handling all-missing training columns, rather than fitting site-specific feature matrices. The 2012 mortality cohort [49] was retained only as an ingestion and outcome-join check; because it has no deterioration onset, we do not treat its 49 decision points as a same-task external validation.

4.2. Labels, Partitions and Eligibility

The supplied 2019 label becomes positive six hours before the challenge-defined clinical onset. We set `label_lead_hours`= 6 to recover the candidate clinical time and `horizon`= 6 to label the six preceding hours positive. Timesteps at or after onset, padded steps, and all steps of a left-censored record are excluded from discrimination and calibration targets. An event at or beyond the end of observed follow-up is not counted as a detectable lead-time case.

For each random seed we use a record-disjoint 60/15/10/15 split into training, validation, calibration and test patients. The scorer, preprocessing,

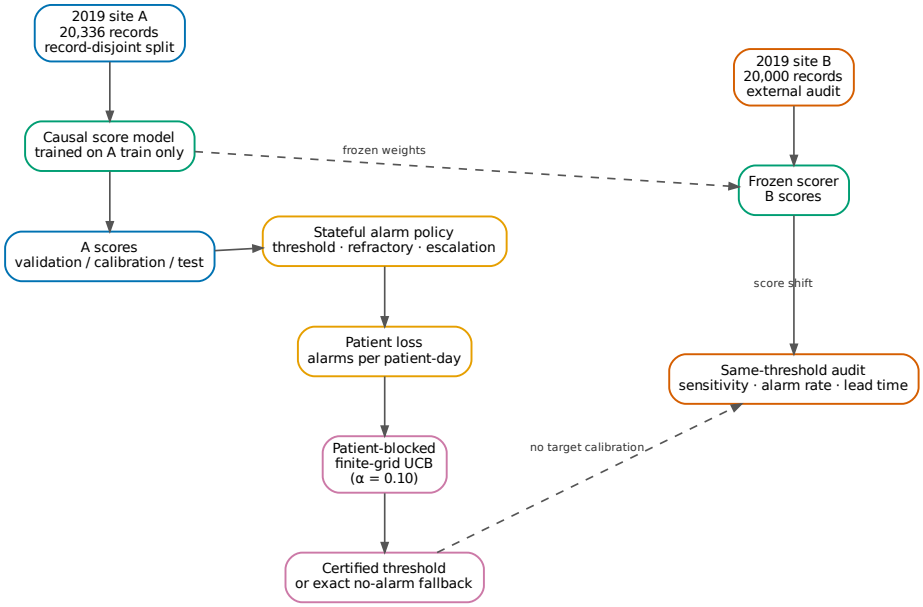


Figure 1: Evaluation design. A causal score model is fitted on site A training records and frozen before site B is touched. The stateful policy replays each patient trajectory to obtain alarms per patient-day; a patient-blocked finite-grid upper confidence bound selects a threshold, or returns the exact no-alarm fallback when no candidate is certified. The primary external audit applies the same threshold without target-site calibration; the target-calibration diagnostic in the Results is a separate oracle analysis.

structural policy parameters and threshold grid are fixed using training and validation records. The calibration partition selects the deployed threshold, and the test partition is touched only for evaluation. Site B is evaluated with the A-trained scorer and the A-calibrated threshold; no B labels are used for target calibration. Three seeds (0, 1, 2) are reported for the prefix logistic model, the masked GRU and the E2 budget sweep. The target-site recalibration diagnostic is reported for all three logistic seeds and for one GRU seed, and is labelled as an oracle analysis throughout.

A separate target-site diagnostic asks how much of the external loss is recoverable when target labels are available. We split the eligible B records into patient-disjoint calibration and test halves of equal size using a site-specific random permutation (seed $1000+s$ for source seed s), recalibrate the

same policy family and threshold grid on B calibration patients, and evaluate on B test patients. This analysis uses target labels for threshold selection and is therefore an oracle upper bound rather than a deployable label-free procedure; it is reported separately from the primary external audit and never used to tune the A model.

The support bound requires an independently declared minimum exposure. We use $H_{\min} = 24$ hours, which retains 15,837 of 20,336 A records and 14,824 of 20,000 B records. This is an eligibility restriction, not a post-hoc filter chosen to improve results. Alarm-rate evaluation counts the full observed exposure, including time after onset; lead-time evaluation uses only the pre-onset window. With hourly decisions, refractory $\tau = 4$ and no escalation, the resulting support bound is $L_{\max} = 144/26 \approx 5.54$ alarms per patient-day.

4.3. Score Models

The primary scorer is a causal prefix-feature logistic regression. At each decision step it uses the last observed value of every variable, the cumulative observation fraction, and admission static features. All features are computed from current and past observations only. Median imputation is fitted on the training partition with empty features retained, so a variable that is entirely missing at site A remains a constant column at site B instead of silently changing the feature dimension. The classifier uses $C = 1$ and a maximum of 2,000 iterations. A unidirectional masked GRU with 64 hidden units is used as a secondary architecture check. It consumes observed values, observation masks, log-transformed elapsed time and static features; it is a reference implementation, not a GRU-D reproduction. The GRU is trained for 20 epochs with Adam at 10^{-3}, batch size 64, and a fixed random seed on an RTX 4090. No LightGBM results are reported because the optional dependency was not installed in the execution environment.

4.4. Alarm Policy and Calibration

The policy family is the stateful rule in Equation (1). We fix refractory $\tau = 4$ decision steps, escalation $\kappa = 0$, and a 12-hour lead window $W = 12$. The candidate grid is 200 equally spaced thresholds from 0.01 to 0.99; the grid is fixed before calibration. The risk level is $\alpha = 0.10$, and the reported certificates use the empirical-Bernstein radius in Equation (7). The selector deploys the smallest threshold whose raw empirical loss plus radius is at most

the requested budget. If no threshold is certified, the exact no-alarm fallback is reported rather than silently substituting the largest finite threshold.

For each patient and threshold, the full stateful trajectory is replayed once to obtain alarms per patient-day. This loss matrix is reused across budgets and across the blocked and timestep procedures, so differences between procedures come only from the confidence radius and not from a different point estimator. The deliberately invalid timestep comparator substitutes calibration timesteps for calibration patients in the Hoeffding radius; it is a diagnostic, not a proposed method.

4.5. Metrics and Uncertainty

Discrimination is summarized by AUROC and AUPRC over usable pre-onset timesteps. Because the positive-timestep prevalence is only 1.27% in the A test partition and 0.72% at B, AUPRC is interpreted against that prevalence rather than against 0.5. Calibration is summarized by 10-bin expected calibration error (ECE); ECE is descriptive and does not establish clinical calibration. Operational metrics are the realised patient-average alarm rate, event sensitivity in the 12-hour lead window, median lead time among detected events, and the fraction of test patients whose per-day alarm count exceeds the requested budget.

Confidence intervals for score metrics use a patient-level bootstrap with 400 resamples, preserving within-patient dependence. The same resampled patients are used for paired model comparisons, and two-sided bootstrap p-values are reported without claiming family-wise error control. Operational curves report the mean and sample standard deviation over seeds; no interval is attached to the single-seed target-site GRU diagnostic.

4.6. Reproducibility and Compute

All raw records and score caches are kept outside the manuscript repository. The code exposes separate entry points for downloading and checksumming the 2019 release, auditing the official files, fitting and caching E1 scores, sweeping E2 budgets, bootstrap intervals and seed aggregation. The full test suite (130 tests) passes on the execution machine. The GRU uses CUDA but no formal GPU-hour or energy measurement was recorded; runtimes are therefore reported as implementation facts, not as a compute-efficiency claim.

Table 1: Public 2019 cohort audit after downloading and checksum verification. Percentages for observed variables are the fraction of records with at least one non-missing value. The two sites are separate hospital systems.

Quantity	Site A	Site B
Records	20,336	20,000
Positive records	1,790	1,142
Left-censored records	203	223
Identifiable pre-onset events	1,587	919
Median observed stay (h)	39	38
$EtCO_2$ observed (%)	0.0	7.6
Temperature observed (%)	33.8	33.9
Lactate observed (%)	3.4	1.9

5. Results

5.1. Cohort Audit and Score Discrimination

The public cohort is large enough to estimate patient-level alarm rates, but its measurement process is far from uniform. Table 1 shows that site A has a higher positive-record fraction than site B (8.8% versus 5.7%), while both sites contain a non-trivial number of left-censored records whose first label is already positive. The observed-variable table also explains why a naive cross-site model can fail: $EtCO_2$ is entirely absent at A but present in 7.6% of B records, and temperature is observed in only about one third of either cohort. A model that drops an all-missing training column would therefore change its input dimension at the external site; the implementation retains such a column as a constant during imputation and the regression test in the repository checks this case explicitly.

Table 2 reports three-seed score discrimination. The prefix logistic model reaches an A-test AUROC of 0.736 ± 0.013 and AUPRC of 0.034 ± 0.006; the masked GRU reaches 0.743 ± 0.011 and 0.043 ± 0.005. These AUPRC values must be read against the 1.27% positive-timestep prevalence: both models rank positive steps several times better than chance, but the absolute positive predictive signal remains modest. On site B the logistic model falls to 0.676 ± 0.005 AUROC and 0.017 ± 0.000 AUPRC, while the GRU reaches 0.683 ± 0.004 and 0.018 ± 0.002. Ten-bin ECE is small in absolute terms (0.0011 and 0.0017 at A; 0.0014 and 0.0018 at B), but it is descriptive and should not be read as evidence of clinical calibration under shift. The drop

Table 2: Three-seed score discrimination (mean ± sample standard deviation). AUPRC is computed over usable pre-onset timesteps; positive-timestep prevalence is 1.27% at A and 0.72% at B. Ten-bin ECE is reported in the text and is descriptive.

	Site A test		Site B external	
Model	AUROC	AUPRC	AUROC	AUPRC
Prefix logistic	0.736 ± 0.013	0.034 ± 0.006	0.676 ± 0.005	0.0172 ± 0.0002
Masked GRU	0.743 ± 0.011	0.043 ± 0.005	0.683 ± 0.004	0.0183 ± 0.0016

is consistent with multisite sepsis-model evaluations that report degraded transport across hospitals [35, 36], and it is larger in AUPRC than in AUROC because prevalence also changes.

The seed-0 patient bootstrap sharpens the comparison without turning a small numerical difference into a claim. At site A the logistic model has AUROC 0.741 with 95% interval $[0.710, 0.766]$ and AUPRC 0.041 with interval $[0.035, 0.051]$; the GRU has AUROC 0.744 $[0.716, 0.772]$ and AUPRC 0.044 $[0.038, 0.056]$. The paired GRU-minus-logistic difference is 0.004 for AUROC with interval $[-0.023, 0.032]$ and two-sided $p = 0.775$, and 0.003 for AUPRC with interval $[-0.006, 0.014]$ and $p = 0.42$. The GRU is therefore a secondary architecture check, not a demonstrated improvement. At site B the AUROC difference is essentially zero, while the AUPRC difference is 0.003 with interval $[0.0002, 0.0059]$ and $p = 0.05$; this isolated interval is reported as exploratory because it is not corrected across metrics or sites.

5.2. Budgeted Alarm Policies and Cross-Site Transport

Figure 2 and Table 3 report the operational frontier for the prefix logistic scorer. On site A, raising the requested budget from 0.25 to 2 alarms per patient-day increases mean sensitivity from 0.185 ± 0.061 to 0.774 ± 0.032, while the realised alarm rate rises from 0.108 ± 0.012 to 1.366 ± 0.026. The realised rate stays below the requested budget at every operating point, which is the expected consequence of a finite-sample upper bound rather than evidence that the policy is slack by design. The tightest budget has the largest seed-to-seed sensitivity variance, because a small change in the calibrated threshold moves several of the 168 eligible A events across the detection boundary. Median lead time among detected events stays between 8.7 and 10 hours, but this conditional median must be read with the corresponding sensitivity: it describes only the events that were detected.

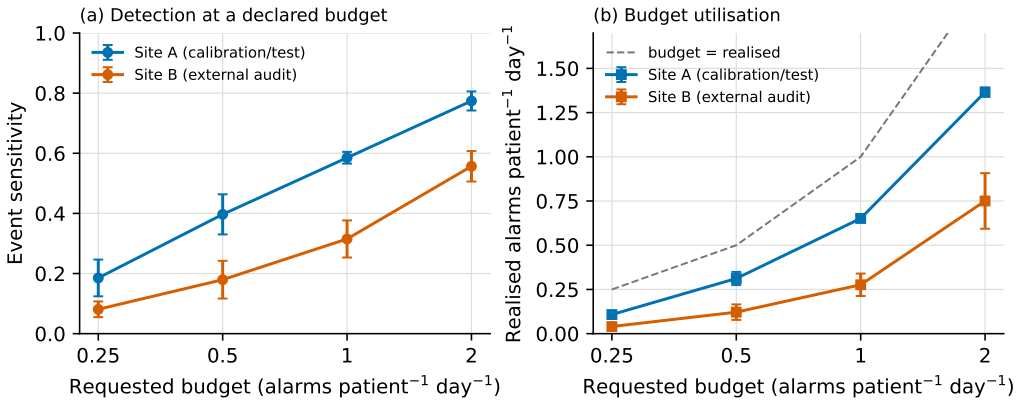


Figure 2: Three-seed E2 operating curves for the prefix logistic scorer. (a) Event sensitivity versus the requested patient-average alarm budget. (b) Realised alarm rate versus the requested budget; the dashed diagonal marks exact budget utilisation. Error bars are sample standard deviations over seeds 0–2. Site B uses the A-calibrated threshold without target labels.

The external audit exposes a qualitatively different operating point. At the 1-per-day budget the A-calibrated policy emits 0.276 ± 0.063 alarms per B patient-day, less than half the A rate, and detects 0.315 ± 0.062 of eligible B events compared with 0.585 ± 0.020 at A. At the tightest budget, B sensitivity falls to 0.081 ± 0.026 while the realised B alarm rate is only 0.040 ± 0.014. The direction of the rate change is not a free lunch: the same threshold is simply less active on the shifted score distribution, and the lost activity is concentrated on true events. This is exactly the failure mode that a threshold-only transport claim would hide, and it is consistent with recent evidence that score distributions and calibration can shift across hospitals [41, 50]. A label-free target-site adaptation study is therefore a separate deployment step, not an optional refinement.

The no-recalibration audit establishes that the source threshold does not travel, but it does not quantify how much of the loss is recoverable by adaptation. To bound that quantity, we split target B into patient-disjoint calibration and test halves (7,412 patients each; on average 354 calibration events and 337 test events over three seeds), recalibrated the same policy family on B calibration labels, and evaluated it on the B test half. This is an oracle diagnostic: target labels are used for threshold selection, so it is not a deployable label-free procedure and must not be compared with the

Table 3: E2 budget sweep for the prefix logistic scorer, mean ± sample standard deviation over three seeds. Rates are alarms per patient-day; lead time is the median among detected events in hours; violation is the fraction of test patients whose per-day count exceeds the budget. The 4-per-day row is at the threshold grid floor; in the full sweep it coincides with 8 per day.

Site A (calibration/test)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.108 ± 0.012	0.185 ± 0.061	9.5 ± 0.5	0.074 ± 0.001
0.50	0.312 ± 0.037	0.397 ± 0.067	8.7 ± 0.6	0.151 ± 0.014
1.00	0.651 ± 0.022	0.585 ± 0.020	9.3 ± 0.6	0.210 ± 0.003
2.00	1.366 ± 0.026	0.774 ± 0.032	10.0 ± 0.0	0.309 ± 0.004
4.00	2.096 ± 0.026	0.894 ± 0.011	10.0 ± 0.0	0.271 ± 0.012

Site B (external audit)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.040 ± 0.014	0.081 ± 0.026	8.5 ± 0.5	0.030 ± 0.010
0.50	0.122 ± 0.044	0.180 ± 0.063	8.2 ± 0.3	0.065 ± 0.023
1.00	0.276 ± 0.063	0.315 ± 0.062	8.3 ± 0.6	0.094 ± 0.025
2.00	0.750 ± 0.157	0.557 ± 0.051	9.0 ± 0.0	0.163 ± 0.038
4.00	1.473 ± 0.250	0.724 ± 0.049	9.3 ± 0.6	0.131 ± 0.025

primary external audit as if it were one. Table 4 reports the result for the logistic scorer. At 1 alarm per patient-day, target calibration raises mean B sensitivity from 0.294 ± 0.082 to 0.497 ± 0.022 and the realised rate from 0.270 ± 0.065 to 0.616 ± 0.058; at 0.25 per day it raises sensitivity from 0.073 ± 0.026 to 0.218 ± 0.036 at a realised rate of 0.159 ± 0.009. The recovery is real but partial, and it is purchased with a higher alarm load and a higher patient-violation fraction because the target-calibrated threshold is lower. The single-seed GRU diagnostic shows the same direction (0.314 to 0.493 at 1 per day). At 2 per day the target-recalibrated selector hits the threshold-grid floor: sensitivity saturates at 0.724 ± 0.056 while the realised rate rises to 1.466 ± 0.260, so the diagnostic identifies a range where adaptation matters rather than a universal gain.

The deliberately invalid timestep comparator is less conservative at the two tightest budgets. At 0.25 per patient-day it raises the mean A alarm rate from 0.108 ± 0.012 to 0.180 ± 0.021 and sensitivity from 0.185 ± 0.061 to 0.283 ± 0.033, with a patient-violation fraction of 0.113 ± 0.008 versus

Table 4: Target-site recalibration diagnostic for the prefix logistic scorer, mean ± sample standard deviation over three source seeds. The target hospital is split 50/50 into patient-disjoint calibration and test halves; the calibration half uses target labels to select the threshold. Sensitivity and realised alarm rate are evaluated on the target test half (7,412 patients, about 337 events). This is an oracle upper bound, not a deployable label-free procedure.

	No recalibration		Target recalibrated	
Budget	Sens.	Rate	Sens.	Rate
0.25	0.073 ± 0.026	0.039 ± 0.014	0.218 ± 0.036	0.159 ± 0.009
0.50	0.173 ± 0.074	0.118 ± 0.043	0.358 ± 0.031	0.360 ± 0.019
1.00	0.294 ± 0.082	0.270 ± 0.065	0.497 ± 0.022	0.616 ± 0.058
2.00	0.544 ± 0.072	0.744 ± 0.165	0.724 ± 0.056	1.466 ± 0.260

0.073 ± 0.001. At 0.50 per day the corresponding rate is 0.425 ± 0.057 versus 0.312 ± 0.037, sensitivity is 0.470 ± 0.052 versus 0.397 ± 0.067, and violation is 0.196 ± 0.024 versus 0.151 ± 0.014. At 1 per day the comparator remains more active (0.744 ± 0.158 versus 0.651 ± 0.022) and less conservative; at 2 per day the finite grid collapses both procedures to the same threshold. The result is therefore a range-specific rather than a universal ranking, but it is the expected qualitative signature of effective-sample-size inflation: the timestep procedure buys sensitivity by shrinking its guard band and paying for it in per-patient budget violations.

The GRU changes the level of the frontier but not its transport behaviour. At 1 alarm per patient-day its three-seed mean A sensitivity is 0.684 ± 0.006 and its B sensitivity is 0.375 ± 0.078, compared with 0.585 ± 0.020 and 0.315 ± 0.062 for the logistic model; the B alarm rate is also higher (0.328 ± 0.164 versus 0.276 ± 0.063). At 0.25 per day the GRU reaches 0.301 ± 0.082 at A and 0.115 ± 0.027 at B, again above the logistic frontier. The GRU is therefore a secondary architecture check rather than a demonstrated improvement: the seed-level spread is large, no paired architecture test is reported, and the external ordering changes with the operating point. A more expressive scorer can move the internal frontier without repairing the site shift, and it can raise external alarm load while doing so. The calibration procedure, not the backbone alone, determines whether a stated alert budget is met.

Two limitations of the grid are visible in the table. First, the 4-per-day row already sits at the threshold grid floor and, in the full sweep, coincides with 8 per day; the grid stops at 0.01; above roughly 2 alarms per patient-day

the budget constraint is no longer binding and the frontier is truncated by the candidate set. Second, every certified operating point has a realised rate below its requested budget, so the experiment demonstrates conservative risk control rather than tight budget utilisation. Neither observation undermines the certificate, but both matter for engineering deployment: a hospital that needs a precise alarm load may prefer a finer grid or a utility-based selection among certified thresholds, and a hospital that needs a higher sensitivity must accept the corresponding increase in realised alert volume. The clinical interpretation of these rates also depends on local workflow and alarm-fatigue tolerance, which this retrospective study does not measure [7, 8].

6. Discussion

The central empirical finding is that a patient-blocked alarm budget is portable as a *constraint* but not as a *threshold*. Within site A, the calibrated policy moves smoothly from 0.19 sensitivity at 0.25 alarms per patient-day to 0.77 at 2 per day, and the realised rates remain below the requested budgets. When the same A-calibrated threshold is applied to site B, the realised rate falls to roughly half the A rate and sensitivity at 1 per day falls from 0.585 to 0.315. The budget itself is still respected in the sense that B's realised mean is below the A certificate, but the clinical yield is much lower. This distinction matters for engineering practice: a site that imports only a threshold has not imported the operating point that the source site validated.

The comparison with the timestep shortcut clarifies what patient blocking buys. At the tightest budgets the timestep procedure selects a lower threshold and increases both sensitivity and the fraction of patients exceeding the budget. That is a real trade, not a statistical technicality: the method that treats correlated hours as independent observations is more willing to spend a fixed nominal risk budget, and the additional alarms are not free at the patient level. At looser budgets the finite grid collapses the two procedures to the same threshold, so the experiment should not be read as a claim that timestep calibration always fails. The defensible conclusion is narrower and stronger: when the estimand is a patient-average rate, the effective sample size is the number of independent patients, and methods that substitute timesteps can become less conservative without a corresponding patient-level guarantee.

The backbone comparison points in the same direction. The GRU is not a demonstrated improvement over the logistic model on site A: its AUROC gain

is small and its patient-bootstrap interval includes zero, while its AUPRC gain is also uncertain. On site B the GRU has a slightly higher AUROC and AUPRC than logistic in the three-seed mean, but the seed-0 paired AUPRC interval only barely excludes zero; in the E2 sweep its external sensitivity at 1 per day is higher (0.375 versus 0.315) with a much wider seed spread and a higher external alarm rate. A more expressive sequence model can therefore move the internal ranking without resolving the distribution shift that dominates the external result. This is consistent with the broader observation that clinical prediction performance is often more sensitive to cohort and workflow differences than to modest architecture changes [35, 36, 50]. It also argues against presenting a backbone-only AUROC table as evidence of deployability.

For engineering deployment, the results support four concrete practices. First, the minimum observed stay used in the support bound should be declared before looking at calibration or test outcomes, because it changes both eligibility and $L_{\max}$; the 24-hour restriction here removes about 22% of A records and 26% of B records. Second, the no-alarm fallback should remain an explicit, auditable state rather than being replaced by the largest finite threshold, since the latter can still alarm and has no certificate. Third, a target site that has local labels should recalibrate on its own patient blocks before adopting a source threshold, because the external rate and sensitivity can move in opposite directions; our oracle split shows that this recovers part of the sensitivity but not without a higher alarm load, and a label-free version remains an open deployment problem. Fourth, the threshold grid and the realised-rate report should be published together, since a grid floor can make high budgets look identical and hide the fact that the constraint is no longer binding. These practices are operational rather than architectural, and they are compatible with the alert-stewardship literature that treats interruptive alerts as a monitored resource [7, 8, 44].

The study has several limitations that bound its claims. It is retrospective and uses a single public challenge cohort with two sites; the record-disjoint split cannot prove that repeated admissions belong to different people, so the independent-patient assumption remains an assumption. The label is the challenge's sepsis definition, not a adjudicated clinical deterioration event, and left-censored records are excluded rather than imputed. The score models are deliberately simple reference learners, the GRU is not a state-of-the-art architecture, and no label-free target-site adaptation, subgroup analysis, observation-sparsification study or clinical outcome evalua-

tion was performed; the target-site recalibration reported here is an oracle diagnostic that uses target labels and is not a deployment procedure. ECE is low in absolute terms but is computed on a low-prevalence timestep problem and should not be read as clinical calibration. The primary external evaluation has no B calibration set by design, so it measures transport failure rather than the best achievable B performance. The three-seed E2 comparison is complete for the two reference scorers, but the target-site diagnostic for the GRU is single-seed, and the study does not estimate GPU-hours or alert-fatigue outcomes.

These limitations do not overturn the methodological result, but they change what should be claimed. The paper does not show that budgeted calibration improves patient outcomes, reduces mortality or is preferred by clinicians. It shows that a stateful policy can be calibrated against a patient-level alarm budget with a finite-sample bound, that the resulting operating curve is measurable on a public cohort, and that the same threshold does not transport unchanged across hospitals. That is a narrower contribution than a clinical-effectiveness claim, but it is a more honest one, and it identifies the target-calibration experiment that a deployment study would need to run next.

7. Conclusion

This paper treated early warning as a stateful alarm policy whose alert load is a declared design constraint rather than an emergent threshold artefact. We derived a patient-blocked finite-grid upper-confidence procedure for the population mean of patient-level alarms per day, separated it from a deliberately invalid timestep effective-sample-size shortcut, and implemented the complete path from causal scoring through policy replay to auditable operating points. On the public 2019 A/B cohort, the procedure produced a monotone internal budget-sensitivity frontier, with A sensitivity rising from 0.185 at 0.25 alarms per patient-day to 0.774 at 2 per day over three seeds. The same A-calibrated threshold did not transport to site B: at the 1-per-day budget, B sensitivity was 0.315 versus 0.585 at A, even though the realised B alarm rate remained below the A certificate. An oracle target-site recalibration raised B sensitivity to 0.497 but also raised the realised rate to 0.616, showing that adaptation can recover part of the loss only by changing the operating point. The timestep comparator was less conservative at tight budgets and increased patient-level budget violations, while the GRU changed

the internal frontier without removing the cross-site shift. These results support budgeted, patient-blocked calibration as an engineering instrument, but they do not establish clinical effectiveness, alert-fatigue reduction or deployment readiness; label-free target adaptation, subgroup and missingness analyses, prospective outcome evaluation and a finer utility-aware grid remain necessary before any clinical claim.

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